Simple-branch zero order for the common genus-2 model

Formal proof packet

27 October 2026

This is a conditional formal result, not a new target experiment or cost comparison. The editable diagram `branch-zero-order.svg`, machine-readable gate table `branch-zero-order.csv`, Lean source, and compiler output accompany this PDF.

Checked statement

At a rational simple root a of the cubic f, the actual local stalk has $y^2 = (x-a)q$, with q a unit and $\operatorname{ord}(y) = 1$. The new theorem `actual_branch_x_factor_zero_order` assumes $N = (x-a)t$ and B is a unit. It proves

$$N - By = y(yq^{-1}t - B), \qquad \operatorname{ord}(N - By) = 1.$$

The second factor is a unit because yt is a nonunit and B is a unit in the local ring. The result is also recorded as a generic discrete-valuation-ring lemma and an actual-stalk specialization.

The discrete-valuation-ring characterization and unit-times-uniformizer valuation are recorded in the Stacks Project, Lemma 10.119.7. The present packet imports the already checked actual-stalk DVR theorem and proves the conditional order statement; the Stacks lemma alone does not establish the genus-2 divisor-class equation.

Checked dependency

$$y^2 = (x-a)q,\ q \text{ unit},\ \operatorname{ord}(y) = 1$$
$$\Downarrow$$
$$N = (x-a)t,\ B \text{ unit} \Longrightarrow N - By = y(yq^{-1}t - B)$$
$$\Downarrow$$
$$\operatorname{ord}(N - By) = 1$$

Open: all law numerators; global divisor/Pic^0 equation; compiled Rust semantics

Verification and boundary

Direct Lean 4.34.1/Mathlib `d13f23b` compilation of `Common22BranchZeroOrder.lean` exited zero. All three theorem axiom audits list only `propext`, `Classical.choice`, and `Quot.sound`; no `sorryAx`. Source SHA-256:

`5e89be1013b4234938e336634a3a20455d3605e287abcf2c6ecb3e321729f5d9`.

The theorem does not establish that every compiled-law numerator factors as $N = (x-a)t$, or that the corresponding rational function has the required divisor class. The matched-rho costs and frozen 1,260-target rows remain unchanged. The accompanying Markdown names the exact command and source files.

After restoring tracked fixtures omitted by the sparse checkout, the repository-wide release library test exited zero with 4,954 passed, 0 failed, and 122 ignored tests using eight test threads. This test result does not extend the theorem's mathematical scope.